

Customer Segmentation Model Development

Final Project Report

Module 06 — Machine Learning (ML-M06) | Customer Personality Analysis & Customer Segmentation Project

1. Problem Statement

The marketing team needs to understand distinct groups within the customer base to design targeted campaigns, improve retention, and personalize offers. This project develops, evaluates, and compares multiple unsupervised clustering algorithms on the cleaned, feature-engineered Customer Personality Analysis dataset to identify actionable, business-interpretable customer segments.

2. Dataset Description

The input is the feature-engineered dataset produced in Module 05 (2,237 customers, 60 columns), which was verified in Task 1: 0 missing values, all columns numeric, and no unresolved data-quality issues (see Module 5 validation report for the explanation of 183 rows sharing an identical behavioural profile, intentionally kept).

Check	Result
Rows	2,237
Columns (input)	60
Missing Values	0
Non-numeric Columns	0

3. Selected Features (Task 2)

Twelve features covering demographics, spending, engagement, and channel usage were selected for clustering:

Feature	Reason
Customer_Age	Life-stage / demographic segmentation driver
Income	Purchasing power indicator
Total_Spending	Direct measure of customer value
Recency	Recent engagement / activity signal
Customer_Tenure	Loyalty / relationship-length signal
Family_Size	Household-scale indicator
Total_Children	Dependents indicator, affects discretionary spend
Total_Purchases	Overall purchase frequency across channels
Total_Campaign_Acceptance	Marketing receptiveness indicator
NumWebPurchases	Digital channel preference/volume
NumStorePurchases	Physical retail channel preference/volume
NumCatalogPurchases	Catalog channel preference/volume

No feature pair exceeded a correlation of 0.85, so all 12 features were retained without further reduction.

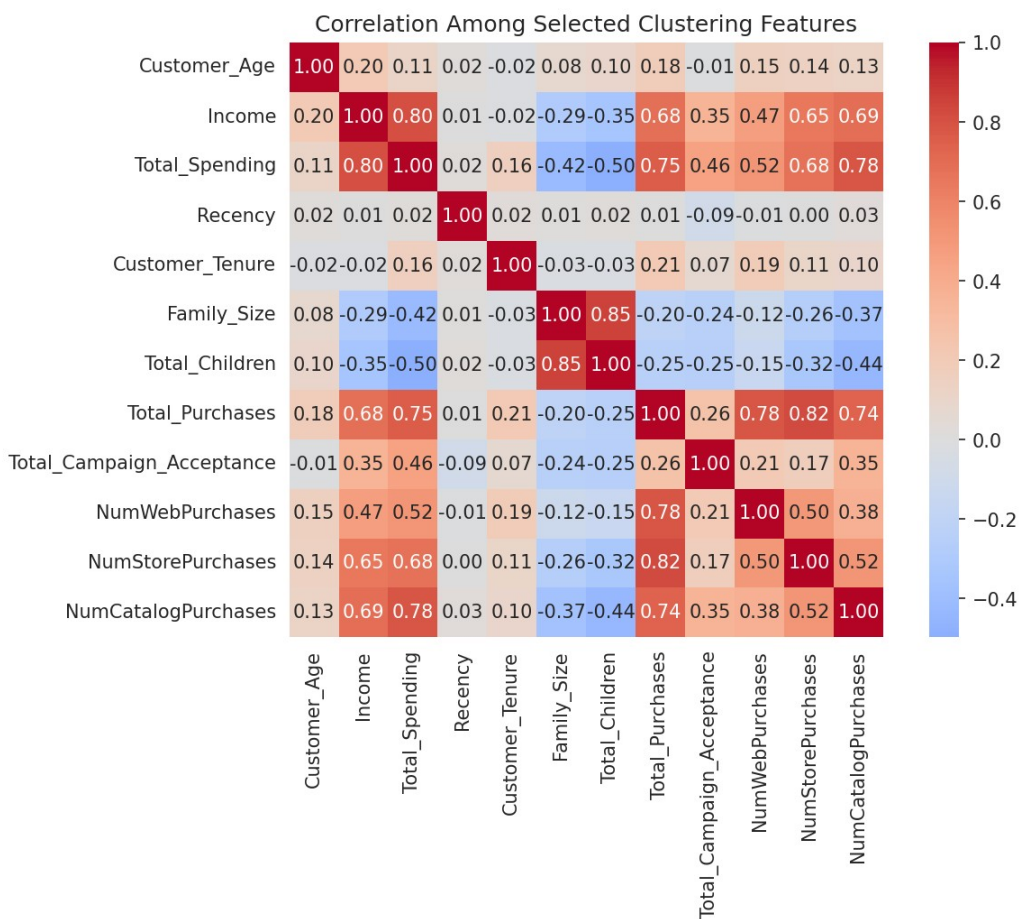


Figure 3.1 — Correlation among selected clustering features.

4. Preprocessing: Feature Scaling (Task 3)

StandardScaler and MinMaxScaler were compared. StandardScaler was selected because distance-based clustering algorithms (K-Means, GMM, Agglomerative) perform best on zero-mean, unit-variance features; MinMaxScaler would let a single extreme value compress the rest of the distribution into a narrow range. The fitted scaler is saved to 04_Models/scaler.pkl for reuse in production.

5. Exploratory Cluster Analysis (Task 4)

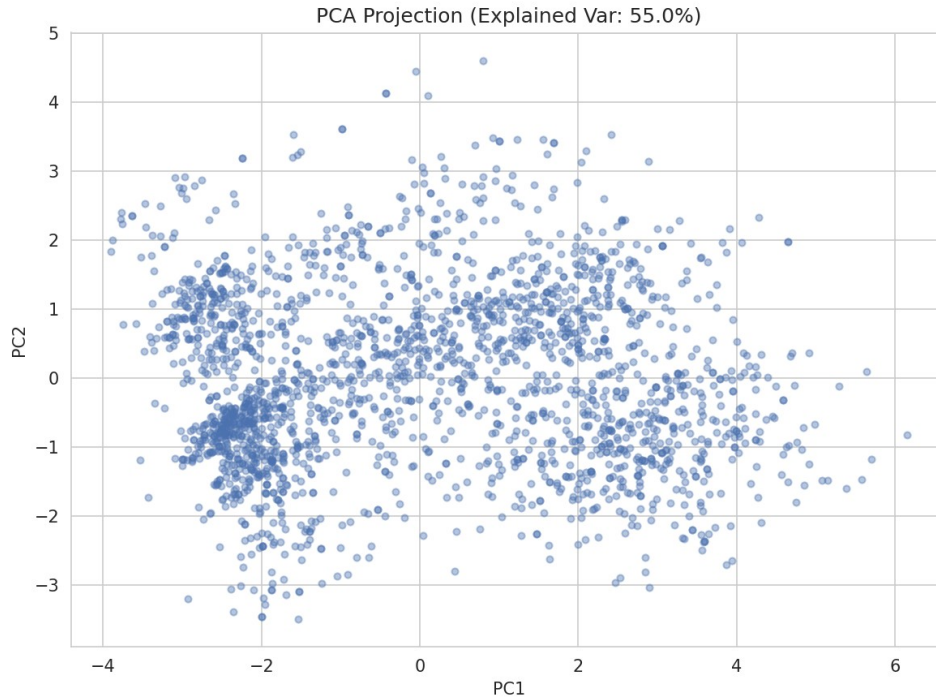


Figure 5.1 — Unlabeled PCA projection of the scaled feature space.

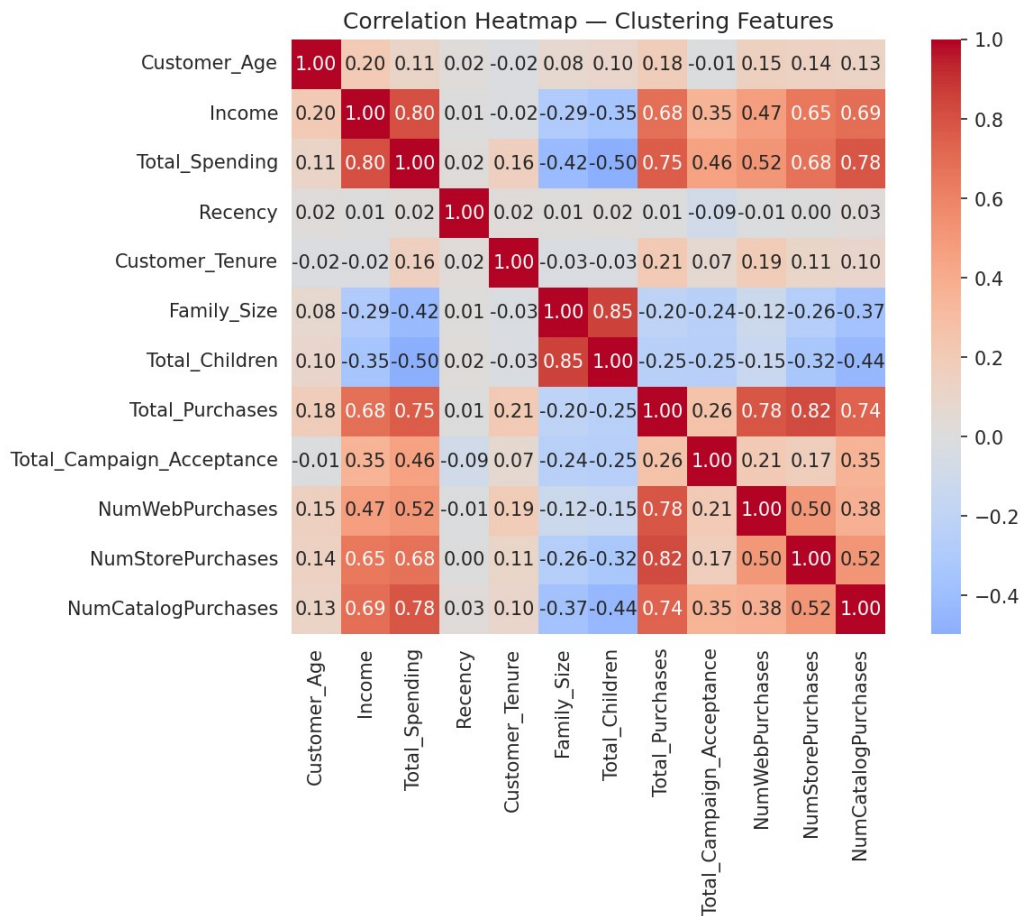


Figure 5.2 — Correlation heatmap of all 12 clustering features.

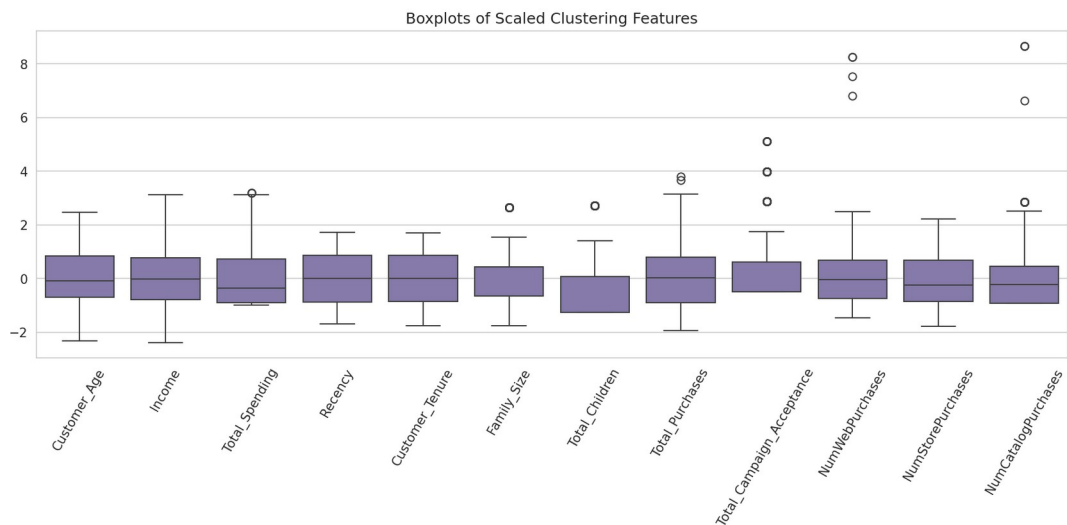


Figure 5.3 — Boxplots of scaled clustering features.

6. Determining the Optimal Number of Clusters (Tasks 5-7)

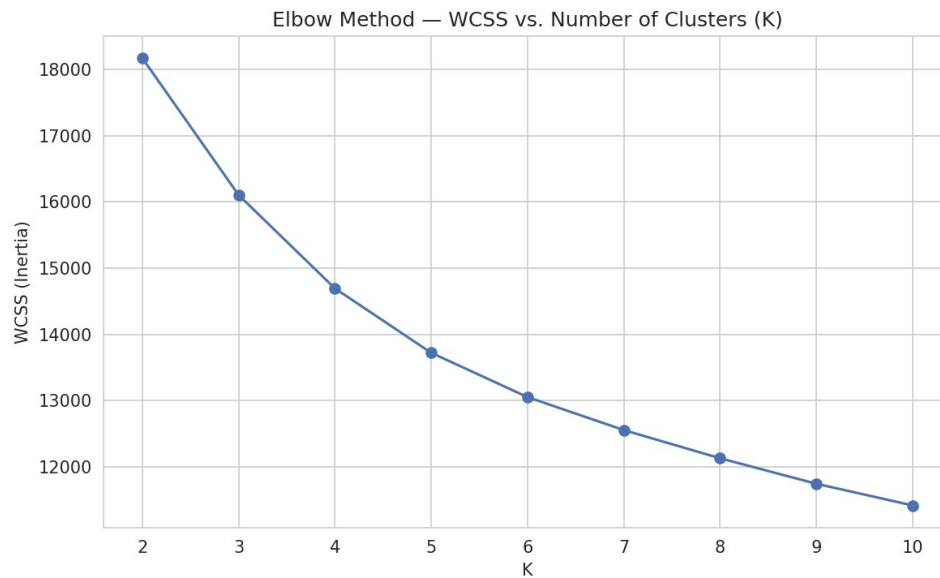


Figure 6.1 — Elbow plot (WCSS vs K). Elbow suggested around K=3.

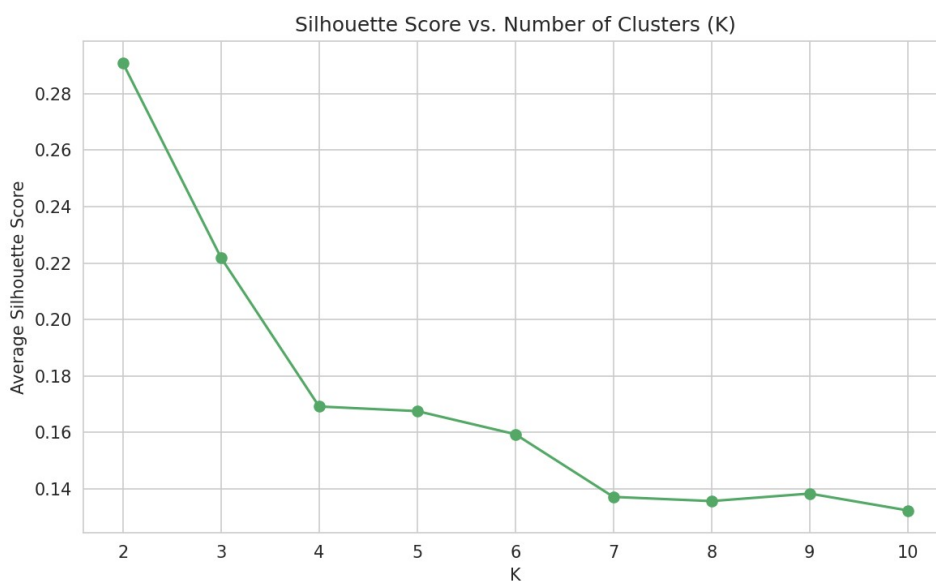


Figure 6.2 — Silhouette score vs K. Statistical peak at K=2.

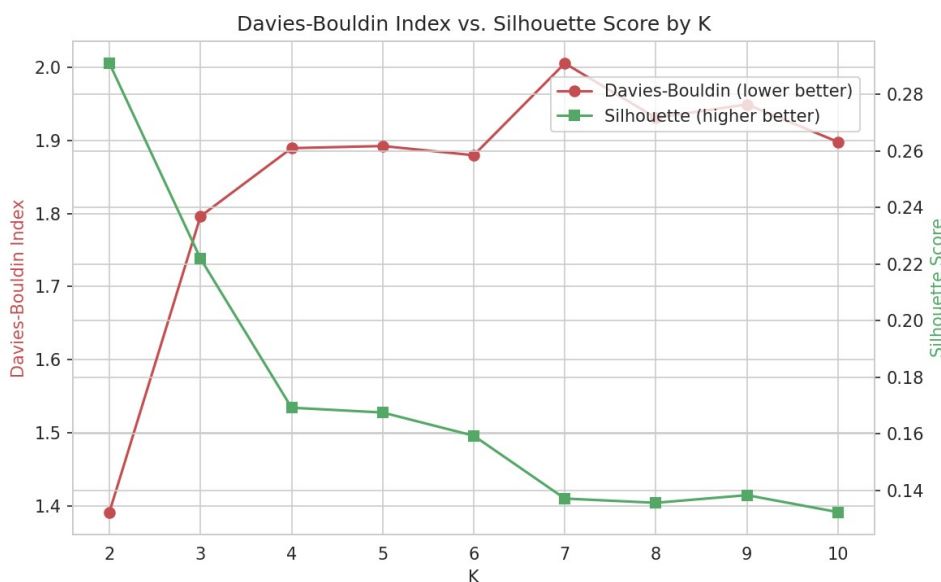


Figure 6.3 — Davies-Bouldin Index vs Silhouette Score by K.

Final K Selection

Silhouette Score and Davies-Bouldin Index both statistically favor K=2, and the elbow plot suggests K=3. However, K=2 is too coarse to produce the actionable, named marketing personas required by the brief (High-Value, Premium, Discount Seekers, At-Risk, New Customers). K=4 was chosen as a documented, business-driven override of the pure statistical optimum: the WCSS improvement visibly flattens from K=4 onward, and the K=4 silhouette score (~0.17) remains a reasonable, positive separation score while unlocking meaningfully richer, business-usable segmentation — a common and defensible practice in applied clustering work.

7. Baseline Model — K-Means (Task 8)

Cluster	Size	% of Total
0	427	19.1%
1	639	28.6%

Cluster	Size	% of Total
2	676	30.2%
3	495	22.1%

Silhouette Score: 0.1692 | Davies-Bouldin Index: 1.8897 | Training time: 0.017s

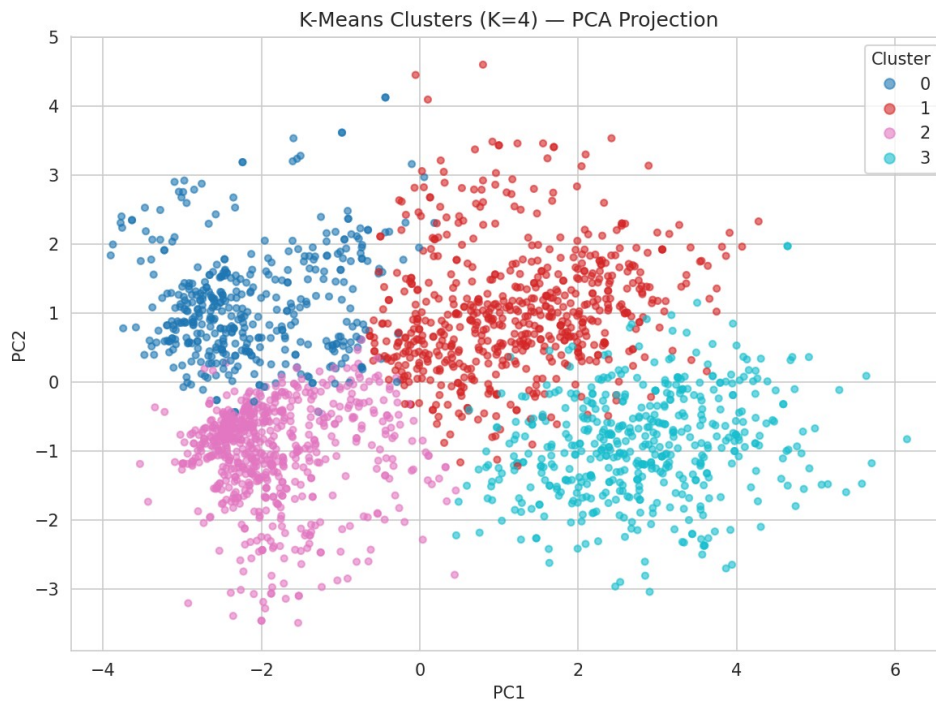


Figure 7.1 — K-Means clusters (K=4), PCA projection.

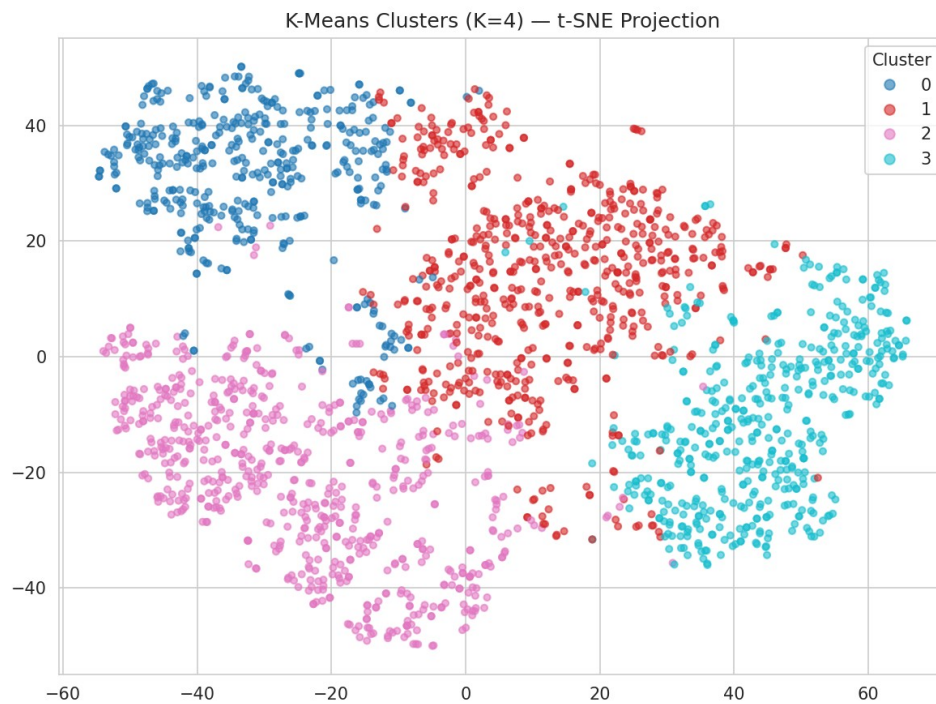


Figure 7.2 — K-Means clusters (K=4), t-SNE projection.

8. Hierarchical Clustering (Task 9)

Agglomerative Clustering (Ward linkage), K=4. Silhouette: 0.1388 | Davies-Bouldin: 1.8047 | Training time: 0.166s.

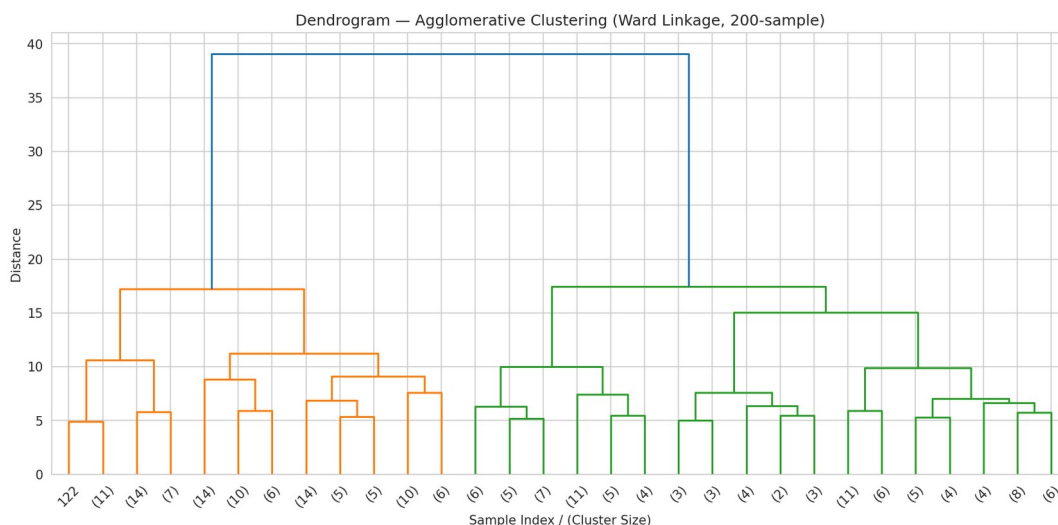


Figure 8.1 — Dendrogram (200-sample, Ward linkage).

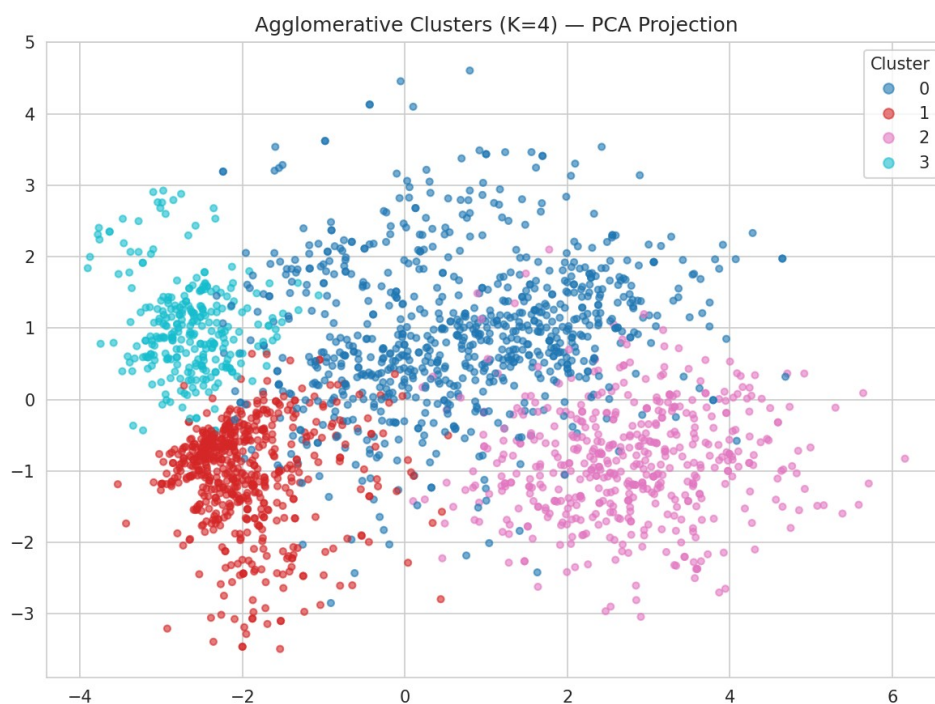


Figure 8.2 — Agglomerative clusters, PCA projection.

9. Gaussian Mixture Model (Task 10)

K=4 components. Silhouette: 0.0476 | Davies-Bouldin: 2.9175 | AIC: 24,916.1 | BIC: 26,989.9 | Training time: 0.094s.
GMM produced the weakest separation of the four algorithms on this feature set.

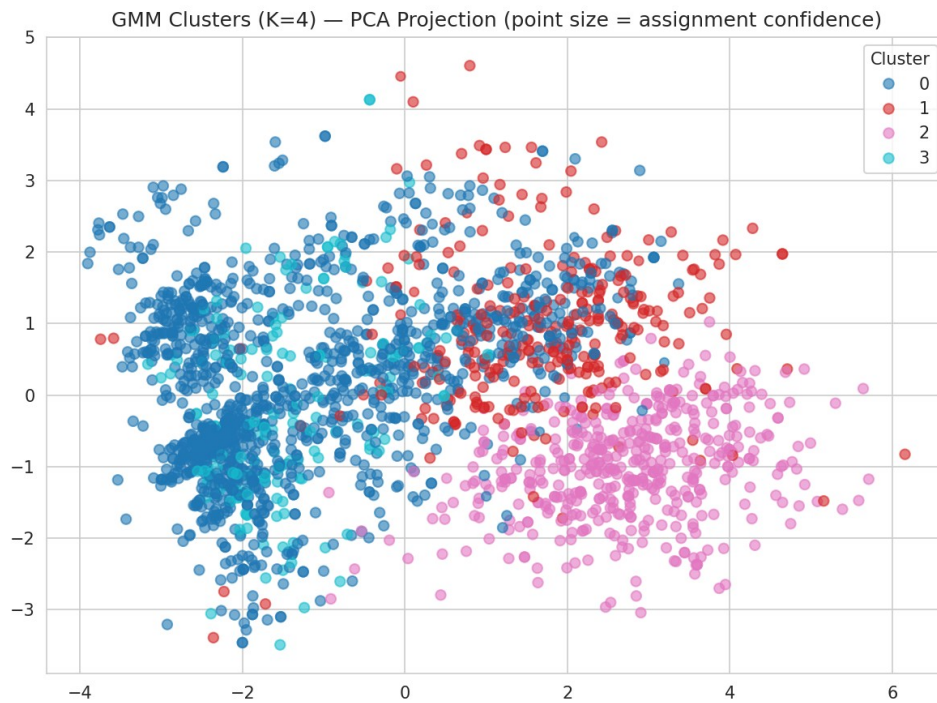


Figure 9.1 — GMM clusters, PCA projection (point size = assignment confidence).

10. DBSCAN Clustering (Task 11)

$\text{eps}=2.709$ (90th percentile of the k-distance plot), $\text{min_samples}=24$ ($2 \times$ feature count). Result: 1 dense cluster with 43 noise points (1.92% of customers) flagged as outliers; density-based parameters did not reveal multiple natural density clusters in this feature space, but did successfully isolate a small set of atypical customers.

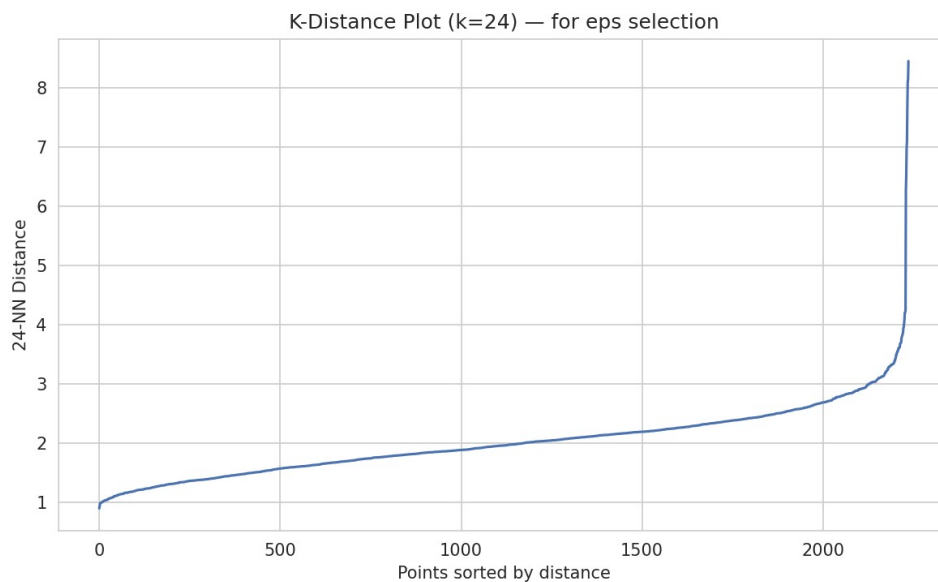


Figure 10.1 — K-distance plot used to select eps.

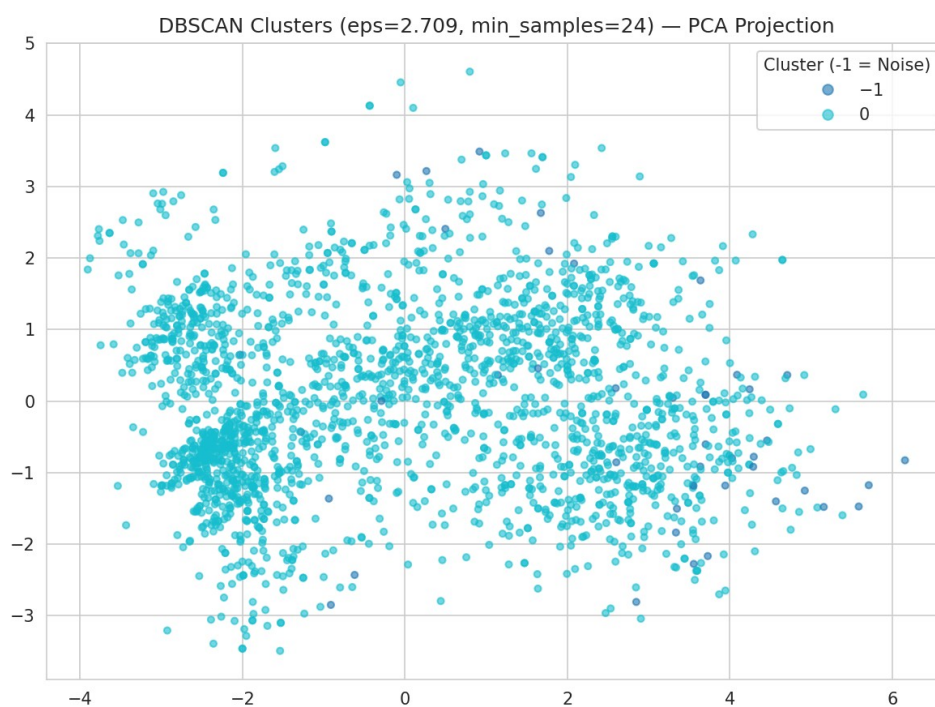


Figure 10.2 — DBSCAN result, PCA projection (-1 = noise).

11. Algorithm Comparison (Task 12)

Algorithm	Silhouette	Davies-Bouldin	Clusters	Time (s)	Interpretability	Handles Noise
K-Means	0.1692	1.8897	4	0.017	High (centroids)	No
Hierarchical	0.1388	1.8047	4	0.166	High (dendrogram)	No
GMM	0.0476	2.9175	4	0.094	Medium (probabilistic)	No
DBSCAN	N/A	N/A	1	0.128	Medium (density-based)	Yes

Recommendation: K-Means achieves the best or near-best silhouette and Davies-Bouldin scores among the comparable-K methods, trains fastest, and produces centroids that are the most directly interpretable for business stakeholders.

12. Cluster Profiling (Task 13)

Cluster	Segment Name	Size	Income	Total Spend	Recency	Family Size	Campaigns Accepted
3	High-Value Customers	495	\$76,705	\$1,430.1	50.1	1.7	1.2
1	Premium / Loyal Buyers	639	\$59,927	\$814.0	47.7	2.8	0.3
0	Discount Seekers / Budget Customers	427	\$42,574	\$131.3	50.2	3.7	0.2
2	New / Developing Customers	676	\$31,891	\$104.9	49.0	2.4	0.2

- High-Value Customers (22.1%): highest income & spend, most campaign-responsive, smallest households — premium offer targets.
- Premium / Loyal Buyers (28.6%): solid income and spend, strong web/store/catalogue activity across all channels.

- Discount Seekers / Budget Customers (19.1%): lower spend, largest families, older average age — value-oriented promotions fit best.
- New / Developing Customers (30.2%): lowest income & spend, youngest average age — nurture with onboarding and welcome offers.

13. Cluster Visualization (Task 14)

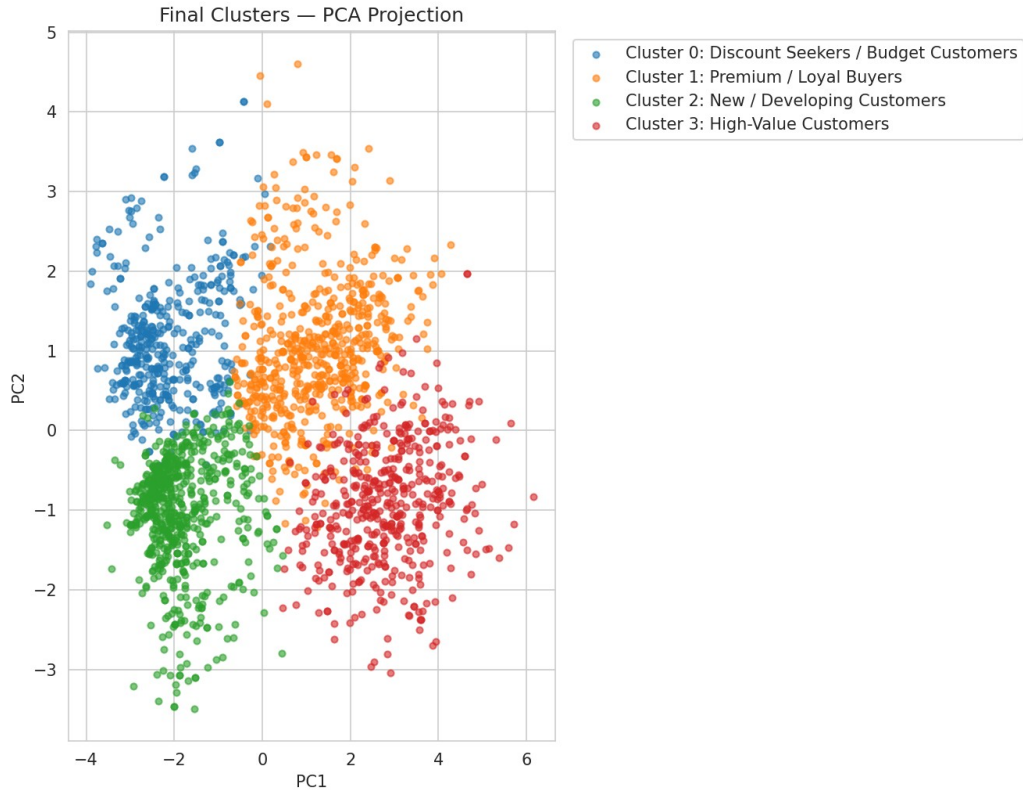


Figure 13.1 — Final named clusters, PCA projection.

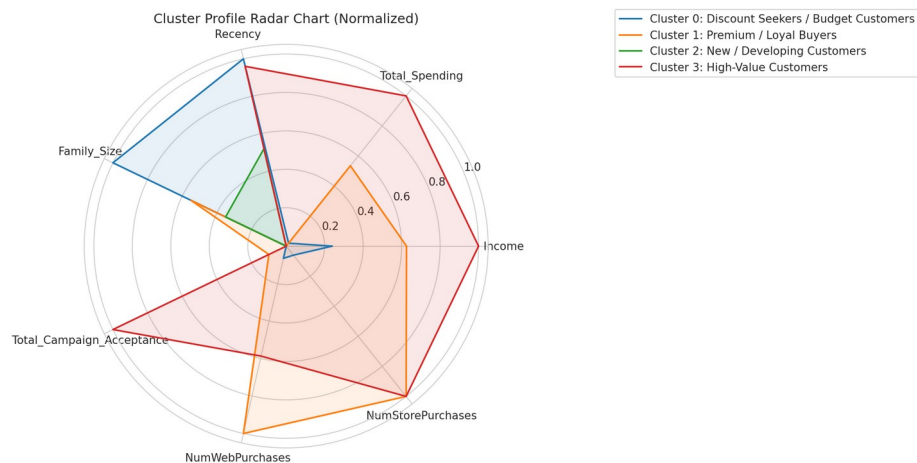


Figure 13.2 — Cluster profile radar chart (normalized features).

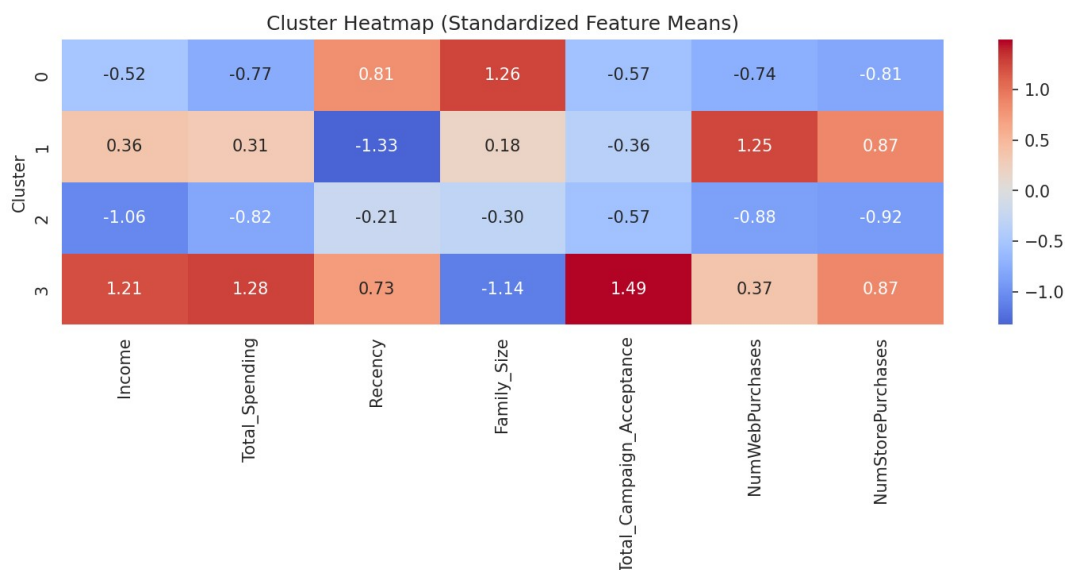


Figure 13.3 — Cluster heatmap (standardized feature means).

14. Cluster Stability Analysis (Task 15)

Test	Score	Interpretation
Random-State Consistency (avg pairwise ARI)	0.9956	Near-perfect agreement across 5 different random seeds.
Bootstrap Sampling Stability (avg ARI)	0.9566	High agreement across bootstrap resamples of the data.

Both stability tests confirm the K=4 K-Means solution is highly reproducible and not an artifact of a particular random initialization or data sample.

15. Final Model Selection (Task 16)

K-Means (K=4) is selected as the final production model, based on:

- Performance: best/near-best Silhouette (0.169) and Davies-Bouldin (1.890) among comparable-K algorithms.
- Business interpretability: cluster centroids translate directly into readable segment-average profiles.
- Stability: ARI > 0.95 across random states and bootstrap resamples.
- Computational efficiency: fastest training time (0.017s), suitable for frequent retraining in production.

16. Production Pipeline (Task 17)

The following artifacts were saved for deployment:

Artifact	File
Feature list	04_Models/production_pipeline.pkl (dict key: feature_list)
Fitted scaler	04_Models/scaler.pkl
Final K-Means model	04_Models/kmeans_model.pkl
Combined production pipeline	04_Models/production_pipeline.pkl / .joblib
Final segmented dataset	05_Final_Deliverables/customer_segments_final.csv

17. Business Recommendations

- High-Value Customers: prioritize premium product bundles, early access, and loyalty perks to protect this high-margin segment.
- Premium / Loyal Buyers: cross-sell campaigns across web, store, and catalogue channels where they are already highly active.
- Discount Seekers / Budget Customers: target with value bundles and family-size promotions rather than premium offers.
- New / Developing Customers: focus on onboarding journeys and welcome incentives to grow spend over time.
- Use the saved production pipeline to score new customers into these four segments automatically as they are acquired.

18. Future Improvements

- Test alternative distance metrics and DBSCAN parameter grids to better capture irregular/outlier customer groups.
- Incorporate temporal/behavioural trend features (e.g. spend trajectory) rather than static snapshots.
- Validate segments against downstream business KPIs (e.g. campaign response uplift per segment) once deployed.
- Periodically retrain the pipeline as customer behaviour evolves, using the saved reusable scaler + model artifacts.